

EDUCATION

Some University

Bachelor of Science in Computer Science (Dean's List)

Graduation May 2025

EXPERIENCE

Amazon, Software Development Engineer | 2025 – Present

- Designed and launched a full-stack AI-powered automation platform (CrewAI) that reduced QA test-plan creation time from 2–3 days to ~30 minutes, handling 100+ automation requests weekly.
- Architected a scalable, loosely coupled cloud deployment on AWS using CI/CD pipelines, ALB, EC2, RDS, and S3, achieving 99.99% uptime and reliable data persistence for 250+ concurrent task executions.
- Developed Amazon QuickSight analyses providing real-time visibility into platform adoption and user engagement, enabling Device OS leadership to drive data-informed Q4 roadmap.
- Implemented a scalable platform analytics infrastructure tracking user behavior, feature adoption, and platform performance, processing 500+ daily events with no impact on application latency.
- Built React frontend with 6+ AI service workflows, implementing dynamic routing, file upload/download capabilities, and real-time task execution monitoring for Device OS QA teams.
- Designed 7+ PostgreSQL tables supporting task lifecycle management, analytics, and user personalization, enabling state tracking, auditability, and consistent platform behavior.

Amazon, Software Development Engineer Intern | May – Aug 2024

- Designed and launched a **microservice on AWS** to automate the end-to-end lifecycle of raw metrics for Alexa family devices, streamlining scheduling, ingestion, processing, and trend visualization.
- Replaced manual anomaly detection with an **RDS-backed solution**, reducing processing time by **90%** and enabling faster identification of faulty devices.
- Architected a **modular codebase** enabling independent integration of storage, algorithms, and visualization components, improving execution efficiency by **96%** and reducing feature rollout time.
- Authored onboarding architecture documentation to accelerate integration of new components and support **agile development cycles**.

FAANG, Software Development Engineer Intern | May – Aug 2023

- Developed a **device memory monitoring framework** for Alexa family devices during test execution, generating logs and visualizations to improve diagnostics and memory analysis.
- Implemented a SQL-based engine for parallel data retrieval from AWS S3, **reducing download time by 80%** and improving large-scale data access efficiency.
- Collaborated with cross-functional teams using **agile methodologies** to gather requirements, design system architecture, and deliver performance-critical features.
- Improved analytics of test execution by adding structured logging for memory usage metrics across devices.

PROJECTS

PayWise | Python, FastAPI, Node.js, Solidity

- Designed and built a biometric-based payment platform enabling secure, low-cost digital payments with blockchain-backed settlement and decentralized storage.
- Implemented an event-driven architecture connecting off-chain identity verification with on-chain execution, enabling **near-instant cross-chain transactions** with minimal user overhead.

LEADERSHIP

Student Government Association, Senator, School of Engineering | 2021 – 2022

- Represented the College of Engineering in student government, advocating for classroom improvements and bridging communication between students and administration resulting in classroom technology modernization.

SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL, HTML, CSS
Frameworks: FastAPI, React, Next.js, SQLAlchemy, CrewAI (agent orchestration), Pandas, NumPy **Cloud & DevOps:** AWS (EC2, RDS, S3, Lambda, IAM, ALB, QuickSight), Docker, Git, GitHub
Concepts: Full-Stack Development, Microservices Architecture, Database Design, API Development, Agile/Scrum